

Find $f(1), f(2), f(3), f(4)$ and $f(5)$ if $f(n)$ is defined recursively by $f(0)=3$ & for $n=0, 1, 2, \dots$

c) $f(n+1) = f(n)^2 - 2f(n) - 2$

Ans: For, $f(1), n=0$

$$\therefore f(1) = f(0)^2 - 2f(0) - 2$$

$$\boxed{f(1) = 1}$$

For, $f(2), n=1$

$$\therefore f(2) = f(1)^2 - 2f(1) - 2$$

$$\boxed{f(2) = -3}$$

For, $f(3), n=2$

$$\therefore f(3) = f(2)^2 - 2f(2) - 2$$

$$= (-3)^2 - 2(-3) - 2$$

$$\boxed{f(3) = 13}$$

For, $f(4), n=3$

$$\therefore f(4) = f(3)^2 - 2f(3) - 2$$

$$= (13)^2 - 2(13) - 2$$

$$\boxed{f(4) = 141}$$

For, $f(5), n=4$

$$\therefore f(5) = f(4)^2 - 2f(4) - 2$$

$$= (141)^2 - 2(141) - 2$$

$$\boxed{f(5) = 19597}$$

$$f(1) = 1, f(2) = -3, f(3) = 13, f(4) = 141, f(5) = 19597$$

d) $f(n+1) = 3^{f(n)/3}$

For $f(1), n=0$

$$\therefore f(1) = 3^{f(0)/3}$$

$$f(1) = 3$$

For, $f(2), n=1$

$$\therefore f(2) = 3^{f(1)/3}$$

$$= 3^{3/3}$$

$$f(2) = 3$$

For, $f(3), n=2$

$$\therefore f(3) = 3^{f(2)/3}$$

$$= 3^{3/3}$$

$$f(3) = 3$$

For $f(4), n=3$

$$f(4) = 3^{f(3)/3}$$

$$= 3^{3/3}$$

$$f(4) = 3$$

For $f(5), n=4$

$$f(5) = 3^{f(4)/3}$$

$$= 3^{3/3}$$

$$f(5) = 3$$

$$f(1) = 3$$

$$f(2) = 3$$

$$f(3) = 3$$

$$f(4) = 3$$

$$f(5) = 3$$

Q find $7^{222} \bmod 11$

$$\Rightarrow 7^{111})^2 \Rightarrow 7^{55})^2 7)^2 \Rightarrow 7^{27})^2 7)^2 7)^2 \Rightarrow 7^{13})^2 7)^2 7)^2 7)^2$$

$$\Rightarrow 7^6)^2 7)^2 7)^2 7)^2 7)^2 \Rightarrow 7^3)^2 7)^2 7)^2 7)^2 7)^2$$

$$\Rightarrow 7^1)^2 7)^2 7)^2 7)^2 7)^2 7)^2 7)^2$$

$$\therefore 7 \bmod 11 = 7$$

$$7^2 \times 7 \bmod 11 = 2$$

$$2^2 \bmod 11 = 4$$

$$4^2 \times 7 \bmod 11 = 1$$

$$1^2 \times 7 \bmod 11 = 7$$

~~$7^2 \bmod 11 = 2$~~
 ~~$50, 7^{222}$~~

$$7^2 \times 7 \bmod 11 = 2$$

$$2^2 \times 7 \bmod 11 = 4$$

$$4^2 \bmod 11 = 5$$

$$\boxed{50, 7^{222} \bmod 11 = 5}$$

$$\boxed{\text{gcd}(123, 277)}$$

$$\begin{aligned} & 277 \bmod 123 \\ & = 31 \end{aligned}$$

$$\text{gcd}(123, 31)$$

$$\begin{aligned} & = 123 \bmod 31 \\ & = 30 \end{aligned}$$

$$\text{gcd}(30, 31)$$

$$\begin{aligned} & = 31 \bmod 30 \\ & = 1 \end{aligned}$$

$$\text{gcd}(30, 1)$$

$$30 \bmod 1$$

$$= 0$$

$$\text{gcd}(123, 277) = 1$$

$$\boxed{\text{gcd}(1529, 14039)}$$

$$\begin{aligned} & = 14039 \bmod 1529 \\ & = 278 \end{aligned}$$

$$\text{gcd}(1529, 278)$$

$$\begin{aligned} & = 1529 \bmod 278 \\ & = 139 \end{aligned}$$

$$\text{gcd}(139, 278)$$

$$\begin{aligned} & = 278 \bmod 139 \\ & = 0 \end{aligned}$$

$$\text{gcd}(1529, 14039) = 139$$

$\boxed{\text{If } n \text{ is an integer and } n^3 + 5 \text{ is odd, then } n \text{ is even.}}$

Let $P \rightarrow n^3 + 5 \text{ is odd}$ & $Q \rightarrow n \text{ is even}$

a) Proof by contraposition:

$$\neg Q \rightarrow \neg P$$

$\rightarrow$ if n is odd then $n^3 + 5$ is even

$$\text{Let, } n = 2k + 1$$

$$n^3 + 5 = (2k + 1)^3 + 5$$

$$= (2k)^3 + 3(2k)^2 \cdot 1 + 3 \cdot 2k \cdot (1)^2 + (1)^3 + 5$$

$$= 8k^3 + 12k^2 + 6k + 1 + 5$$

$$= 8k^3 + 12k^2 + 6k + 6$$

$$= 2(4k^3 + 6k^2 + 3k + 3)$$

$$\text{Let, } 4k^3 + 6k^2 + 3k + 3 = m$$

$$\therefore n^3 + 5 = 2m$$

[proved]

b) Proof by contradiction:

$$P \wedge \neg Q$$

$\Rightarrow n^3 + 5$ is odd and n is odd

$$\therefore n = 2k + 1$$

$$n^3 + 5 = (2k + 1)^3 + 5$$

$$= 8k^3 + 12k^2 + 6k + 1 + 5$$

$$= 2(4k^3 + 6k^2 + 3k + 3)$$

$$\text{Let } 4k^3 + 6k^2 + 3k + 3 = m$$

$$n^3 + 5 = 2m$$

[if n is odd then $n^3 + 5$ is even]

[proved]

Let $Q(x, y)$ be " x has sent an email message to y ",
where x & y both domains consist of all students in class.

a) $\exists x \exists y Q(x, y)$

$\Rightarrow$ At least one student
has sent email to
~~at least~~ another student

b) $\exists x \forall y Q(x, y)$

$\Rightarrow$ There is at least one student
who has sent an email to
everyone in class.

c) $\forall x \exists y Q(x, y)$

$\Rightarrow$ Every student in class
has sent an email to
at least one student.

d) $\exists y \forall x Q(x, y)$

$\Rightarrow$ At least one student has
received an email from
every student in class.

e) $\forall y \exists x Q(x, y)$

$\Rightarrow$ Every student has received
an email from at least
one student.

f) $\forall x \forall y Q(x, y)$

$\Rightarrow$ Every student has
received an email from
every student in class.

☐ Let $L(x, y)$ be "x loves y", where domain for both x and y consists of all people in the world.

a) Everybody Loves Jerry

$$\Rightarrow \forall x L(x, \text{Jerry})$$

b) Everybody loves somebody

$$\Rightarrow \forall x \exists y L(x, y)$$

c) There is somebody who loves everybody.

$$\Rightarrow \exists y \forall x L(x, y)$$

d) Nobody Loves everybody

$$\Rightarrow \neg \exists x \forall y L(x, y)$$

e) There is somebody whom Lydia does not Love.

$$\Rightarrow \exists y \neg L(\text{Lydia}, y)$$

f) There is somebody whom no one Loves.

$$\Rightarrow \exists y \neg \forall x L(x, y)$$

g) There is exactly one person whom everyone loves.

$\Rightarrow$ Let unique person be z

$$\Rightarrow \forall x L(x, z)$$

h) There are exactly two people whom Lynn loves.

$\Rightarrow$ Let two people be, a & b

$$\Rightarrow L(\text{Lynn}, a) \wedge L(\text{Lynn}, b)$$

i) Everyone loves himself or herself

$$\Rightarrow \forall x L(x, x)$$

j) There is someone who loves no one beside himself or herself.

$$\Rightarrow \exists x L(x, x)$$

☐ $C(x)$ is "x is a comedian" & $F(x)$ is "x is funny"
domain consists of all people.

a) $\forall x(C(x) \rightarrow F(x))$
if $\rightarrow$ then all are
 $\rightarrow$ All comedians are funny

b) $\forall x(C(x) \wedge F(x))$
 $\rightarrow$ Everyone is a comedian
and everyone is funny

c) $\exists x(C(x) \rightarrow F(x))$
 $\rightarrow$ There is someone who, if
they are a comedian then is funny

d) $\exists x(C(x) \wedge F(x))$
 $\rightarrow$ There is someone who
is a comedian and is
funny.

☐ Truth value of $(p \rightarrow q) \wedge (\neg p \rightarrow q)$

P	q	$P \rightarrow q$	$\neg P$	$\neg P \rightarrow q$	$(P \rightarrow q) \wedge (\neg P \rightarrow q)$
0	0	1	1	0	0
0	1	1	1	1	1
1	0	0	0	1	0
1	1	1	0	1	1

☐ Truth table of $(p \leftrightarrow q) \vee (\neg p \leftrightarrow q)$

P	q	$\neg P$	$P \leftrightarrow q$	$\neg P \leftrightarrow q$	$(P \leftrightarrow q) \vee (\neg P \leftrightarrow q)$
0	0	1	1	0	1
0	1	1	0	1	1
1	0	0	0	1	1
1	1	0	1	0	1

⊞ Encrypt message "STOP GLOBAL WARMING"
using shift cipher, with shift $k=11$

⇒ "DEZA RWZMLW HLCXTYR"

⊞ There are infinitely many primes.

⇒ Let $Q = P_1 \cdot P_2 \cdot P_3 \cdot \dots \cdot P_n$ [where Q is the complete list of prime numbers]

Now, create a new prime number

$$\Rightarrow Q_n = (P_1 \cdot P_2 \cdot P_3 \cdot \dots \cdot P_n) + 1$$

∴ In this new prime there is a remainder of 1.

∴ But in the original list there was no remainder.

So, Q_n itself is a new prime number

[proving the statement true]

⊞ If n is a composite integer, then n has a prime divider less than or equal to $\sqrt{n}$.

⇒ Using Trial division,

∴ n is composite,

∴ then $n = a \times b$

⇒ Using proof by contradiction,

∴ suppose both factors are greater than $\sqrt{n}$

$$\Rightarrow a > \sqrt{n}, b > \sqrt{n}$$

$$\Rightarrow a \cdot b > \sqrt{n} \times \sqrt{n}$$

$$\Rightarrow ab > n$$

[But this is impossible, because $ab = n$]

[So statement is true]

⊞ Using Linear search

$a = [1, 3, 4, 5, 6, 8, 9, 11]$

find := 9

location := 0

For $i = 0$ to n do

if $a[i] = \text{find}$ then

location := i

end if

end for

return location

⊞ Using Binary search

array = [1, 2, 3, 4, 5, 6, 8, 9, 11]

find := 9

low := 0

high := length(array) - 1

while low <= high

middle := Floor((low + high) / 2)

if array[middle] < find

low := middle + 1

else if array[middle] > find

high := middle - 1

else

print return "Found in index", middle

stop

end if

end while

Print "not found"

☐ $a = [6, 2, 3, 1, 5, 4]$

for $i = 1$ to $a-1$

for $j = 1$ to $a-1$

if $a_j > a_{j+1}$, then interchange a_j and a_{j+1}

end if

end for

end for

print a